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Independent Study
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WEEKLY PROGRESS REPORT

My GitHub Repo: <https://github.com/akuikel/detectingAIGeneratedCode>

Weeks 1 & 2

Project Overview

- **Project Title:** Extending AI-Generated Source Code Detection to Next-Generation LLMs
<https://github.com/JaeWorld/ICSE2025-AI-Detector>
<https://arxiv.org/abs/2411.04299>
- **Based On:** ICSE 2025 Paper: arXiv:2411.04299
- **Advisor:** Dr. Zhang

Overview

The goal of this independent study is to extend the ICSE 2025 paper "*An Empirical Study on Automatically Detecting AI-Generated Source Code: How Far Are We?*" by evaluating newer large language models not tested in the original work. Weeks 1 and 2 focused entirely on understanding the foundational paper and setting up the research environment.

Week 1 — Literature Review and Paper Understanding

Activities Completed:

- Read the full paper (arXiv:2411.04299) from ICSE 2025 by Suh et al. (UC Irvine)
- Studied the three detection approaches: LLM-based classification, ML with static code metrics, and ML with code embeddings
- Reviewed the paper's dataset structure: MBPP, HumanEval-X, and CodeSearchNet across Python, Java, and C++

- Analyzed the four LLMs tested in the paper: ChatGPT, GPT-4, Gemini Pro, and Starcoder2-Instruct
- Reviewed the results table in Results.xlsx from the original GitHub repository
- Understood the ablation study comparing code-only, AST-only, and code+AST combined embeddings

Key Findings from the Paper:

- Existing AIGC detectors (GPTZero, Sapling, DetectGPT) perform poorly on source code — they were built for natural language
- GPTSniffer (previous best method) does not generalize across programming languages or LLMs other than ChatGPT
- The best approach — CodeT5+ embeddings with logistic regression — achieves $F1 = 0.8255$
- AST features improve detection: combined code+AST embeddings outperform code-only embeddings
- Different LLMs produce differently detectable code, suggesting model-specific stylistic fingerprints

Week 2 — Gap Analysis and Research Planning

Activities Completed:

- Cloned the original GitHub repository (JaeWorld/ICSE2025-AI-Detector) and examined all source files
- Analyzed the uploaded .pkl files (programs.pkl, programs_bak.pkl) to understand the actual data format used in the pipeline
- Performed a systematic gap analysis comparing what the paper claims vs. what exists in the repository
- Identified that the repository is missing ~25 of 28 dataset slices (only 3 Java/HumanEval-X CSVs are present)
- Discovered a missing script: the CodeT5+ embedding generator is not included in the original repo
- Examined all three pipeline scripts: test_embedding.py, llm_zero_shot.py, ast-generator.py
- Formulated the research extension plan: add Claude, GPT-4o, and Gemini 2.5 as new LLMs

- Created new GitHub repository (akuikel/detectingAIGeneratedCode) for this extension project

Research Extension Plan Finalized:

- **Dataset:** Use all 164 Python problems from HumanEval-X (same as paper's primary benchmark)
- **New LLMs:** Claude Haiku (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google)
- **Approach:** Replicate the best-performing approach from the paper (embedding + ML classifier)
- **Embedding model:** Use CodeBERT (same model as GPTSniffer baseline) due to Python 3.13 compatibility constraints
- **Evaluation:** 5-fold stratified cross-validation, report F1, Human F1, AI F1, and Accuracy

Planned Next Steps (Week 3)

- Obtain API keys for Anthropic, OpenAI, and Google
- Write code generation script to query all 3 LLMs for 164 solutions each
- Build data preparation and merging pipeline
- Begin AST generation using tree-sitter